

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: Data Visualization with Power BI and Tableau
PRACTICAL NO. 4

AIM: Practical 4: Performing Data Analysis using Calculations in Tableau

- a) Create basic calculated fields using formulas
- b) Perform row-level calculations
- c) Perform aggregate-level calculations
- d) Implement FIXED Level of Detail (LOD) calculations
- e) Use INCLUDE and EXCLUDE LOD expressions
- f) Create and use parameters for dynamic analysis
- g) Apply calculations to fix data issues and enhance insights

a) Create basic calculated fields using formulas

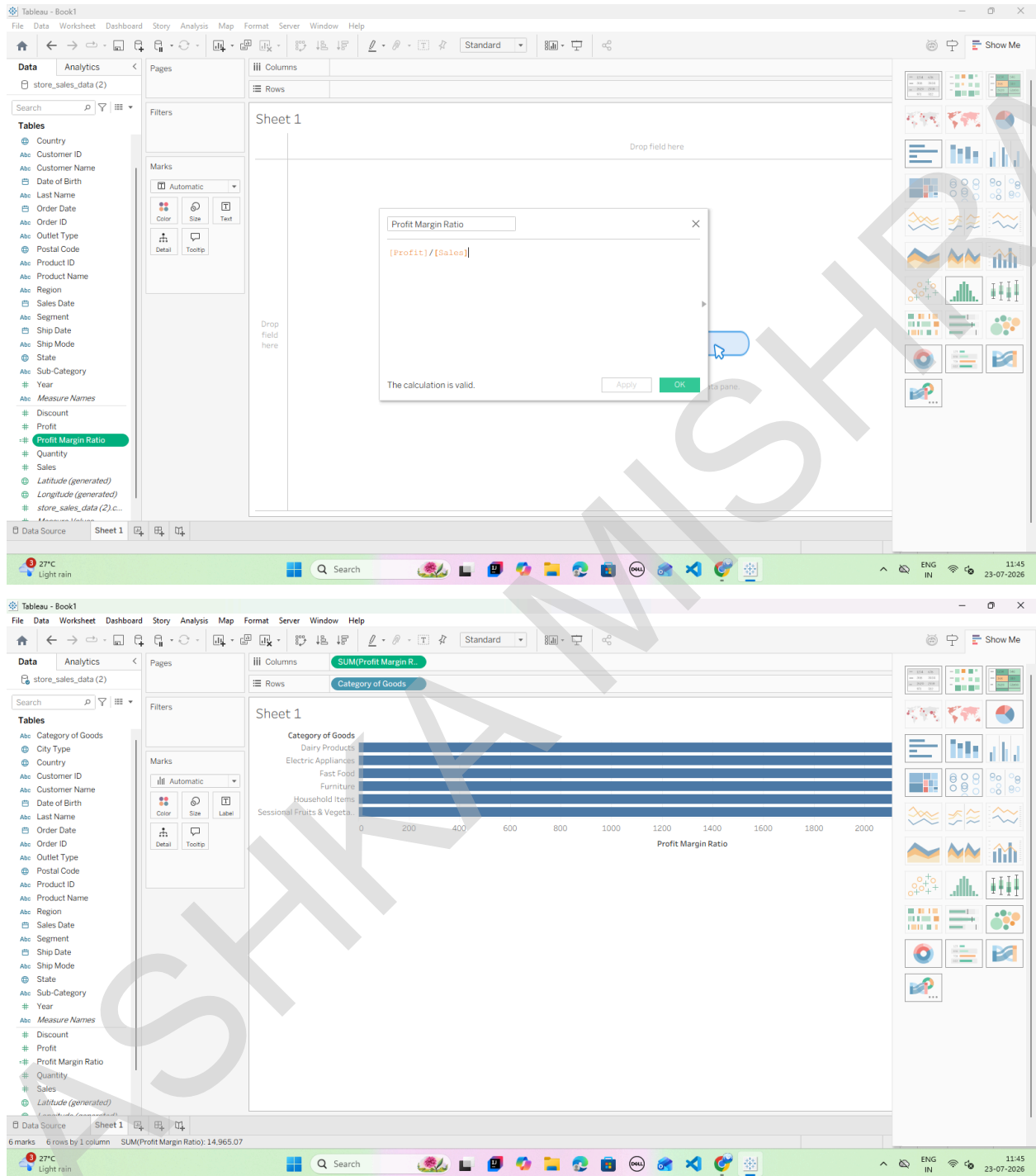
1. Connecting to the Dataset

The screenshot shows the Tableau Desktop interface with the 'store_sales_data (2)' dataset connected. The 'Connections' pane on the left shows the dataset as a 'Text file'. The 'Files' pane lists several CSV files, including 'startup_funding.csv', 'store_sales_data (2).csv', and 'Superstore.csv'. The main view displays a table with columns: Customer ID, Customer Name, Last Name, Date of Birth, Sales, Year, and Outlet Type. The 'Fields' section on the left lists various fields like Customer ID, Customer Name, Last Name, Date of Birth, Sales, Year, Outlet Type, City Type, Category of Goods, Region, Country, and Segment.

Customer ID	Customer Name	Last Name	Date of Birth	Sales	Year	Outlet Type
CUST000001	Curtis	Krause	29-11-1973	3,344.69	2023	Large
CUST000002	Jeremy	Foster	30-08-1987	37,094.41	2022	Medium
CUST000003	Ariel	Mccann	15-12-1980	35,501.08	2023	Large
CUST000004	Colleen	Maynard	23-05-1989	8,969.33	2021	Medium
CUST000005	Gina	Holden	31-12-1957	35,530.44	2021	Small
CUST000006	Danny	Stevenson	25-02-1967	420.76	2019	Large
CUST000007	Stephanie	Mccoy	19-10-2003	31,588.07	2022	Medium
CUST000008	Henry	Cruz	22-05-2013	4,419.87	2021	Small
CUST000009	Kristen	Lutz	08-07-2005	41,494.09	2023	Large
CUST000010	Gregory	Cuevas	10-04-1981	39,717.64	2022	Medium
CUST000011	Sara	Yang	29-04-1956	42,842.38	2019	Medium
CUST000012	Kenneth	Johnson	12-04-1954	1,036.80	2022	Small
CUST000013	Charles	Krueger	02-07-2004	39,274.36	2023	Small
CUST000014	Brenda	Campbell	26-03-1952	31,690.81	2023	Small
CUST000015	Michael	Smith	24-07-1964	40,712.36	2022	Large

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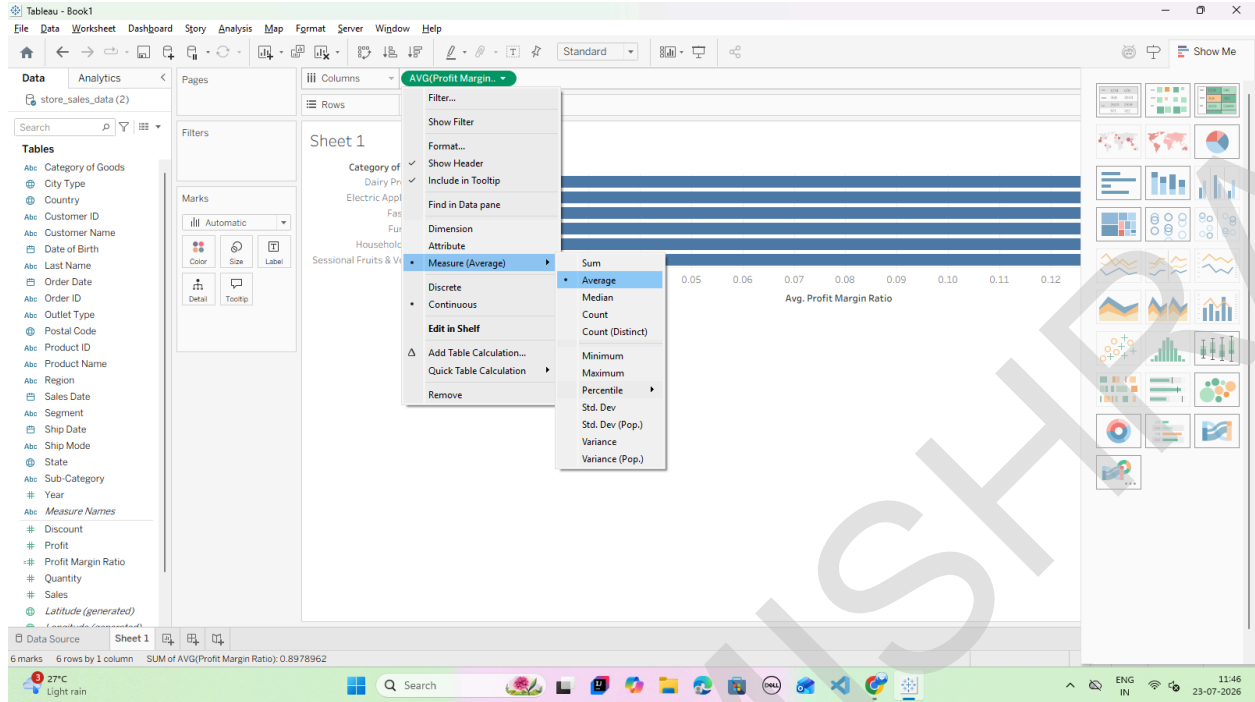
2. Building a Basic Custom Calculation



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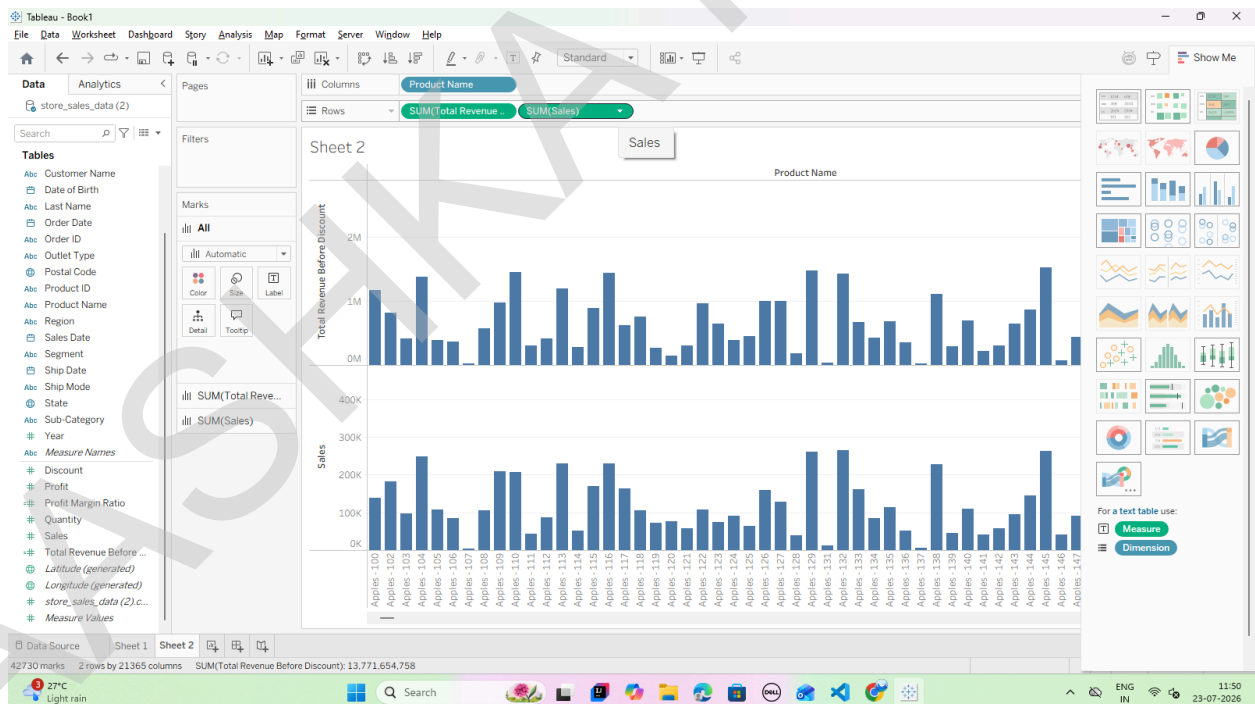
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b) Perform row-level calculations

1. Creating a Row-Level Calculated Field

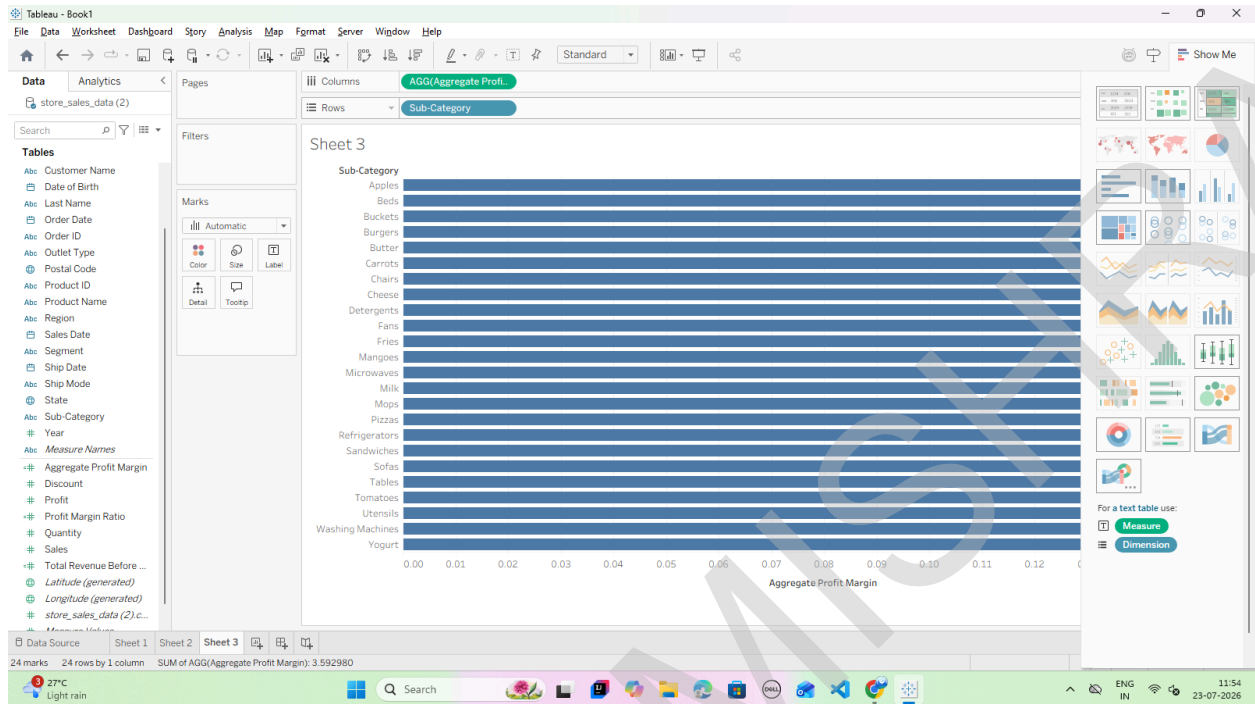


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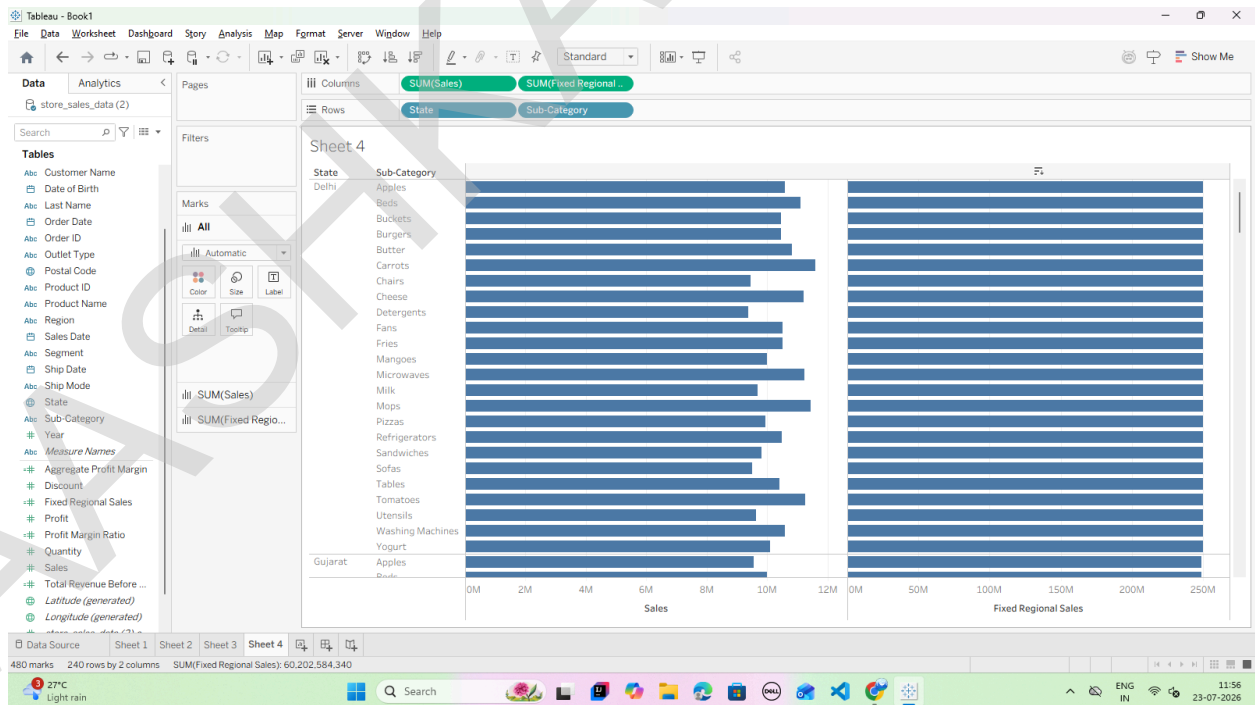
c) Perform aggregate-level calculations

1. Creating an Aggregate Calculated Field



d) Implement FIXED Level of Detail (LOD) calculations

1. Creating a FIXED LOD Calculated Field

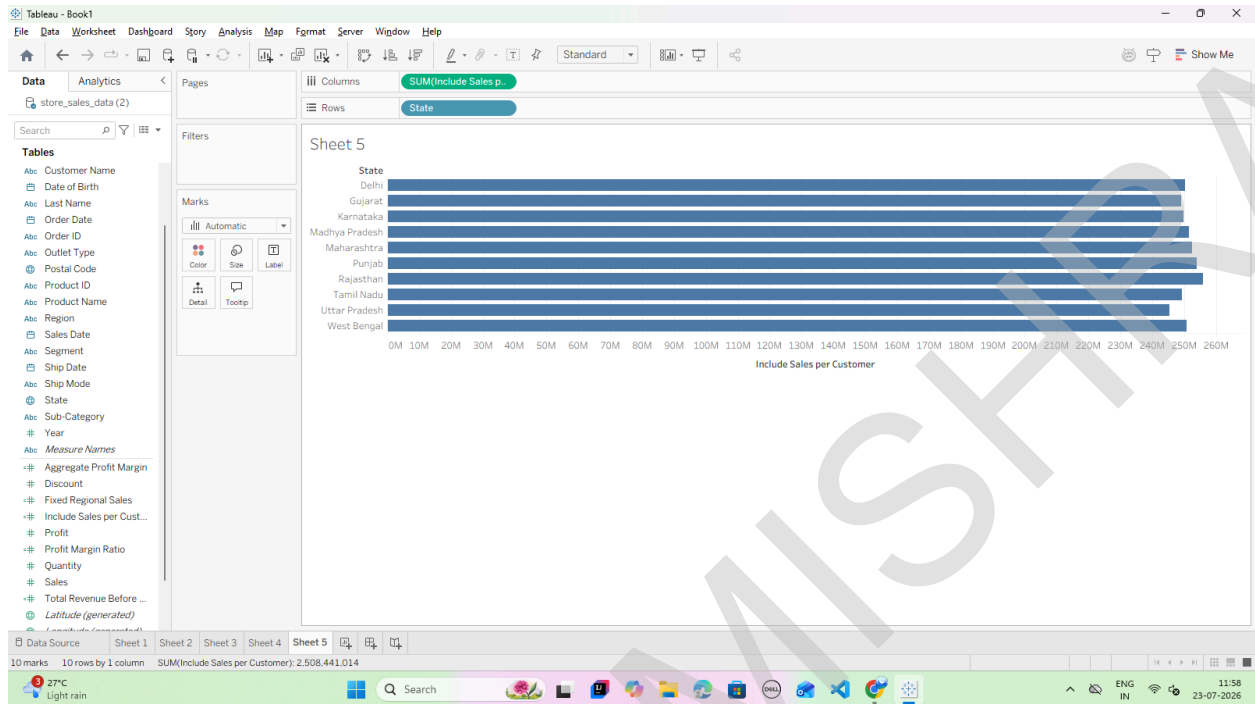


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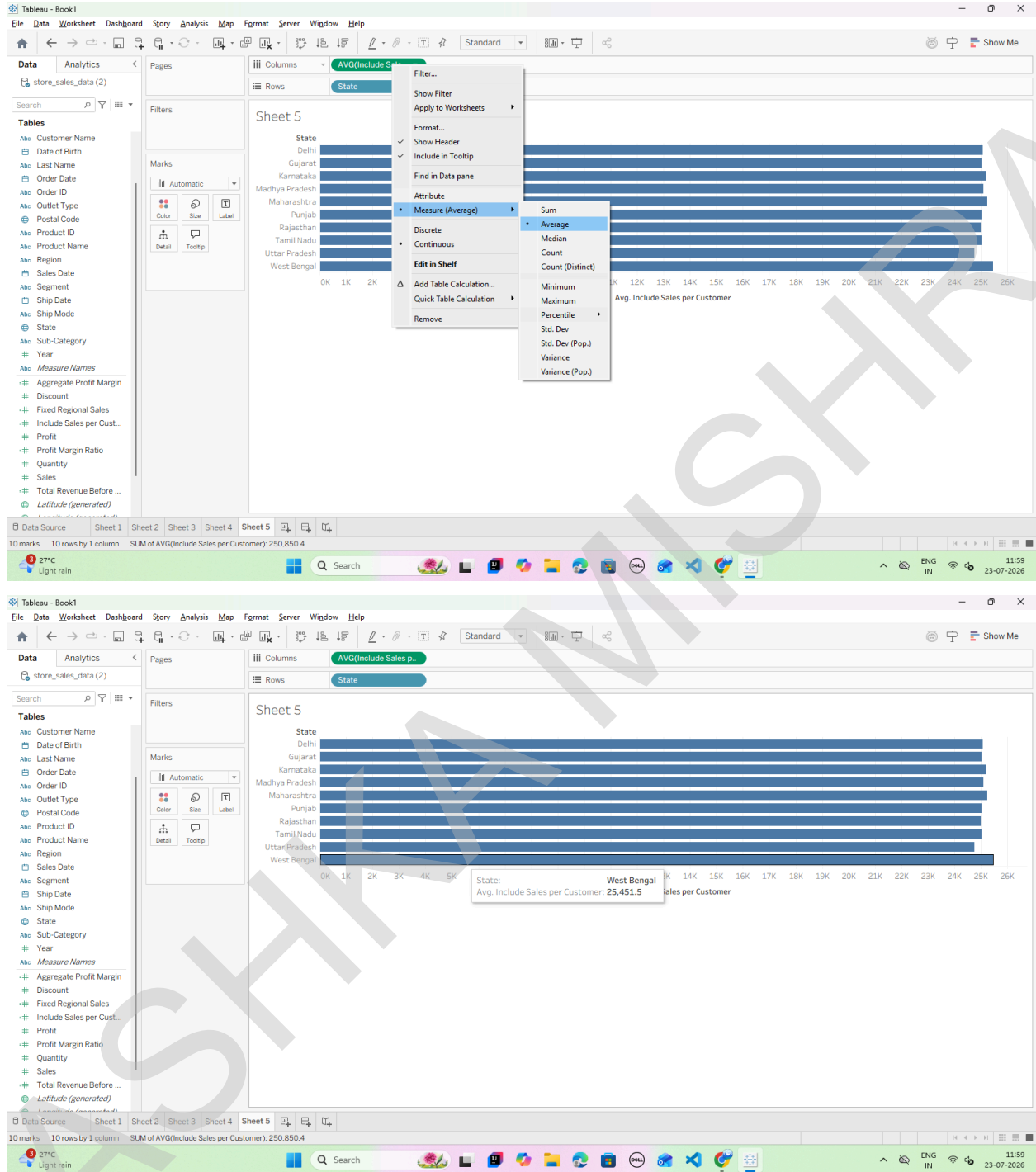
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e) Use INCLUDE and EXCLUDE LOD expressions

1. Creating an INCLUDE LOD Calculated Field



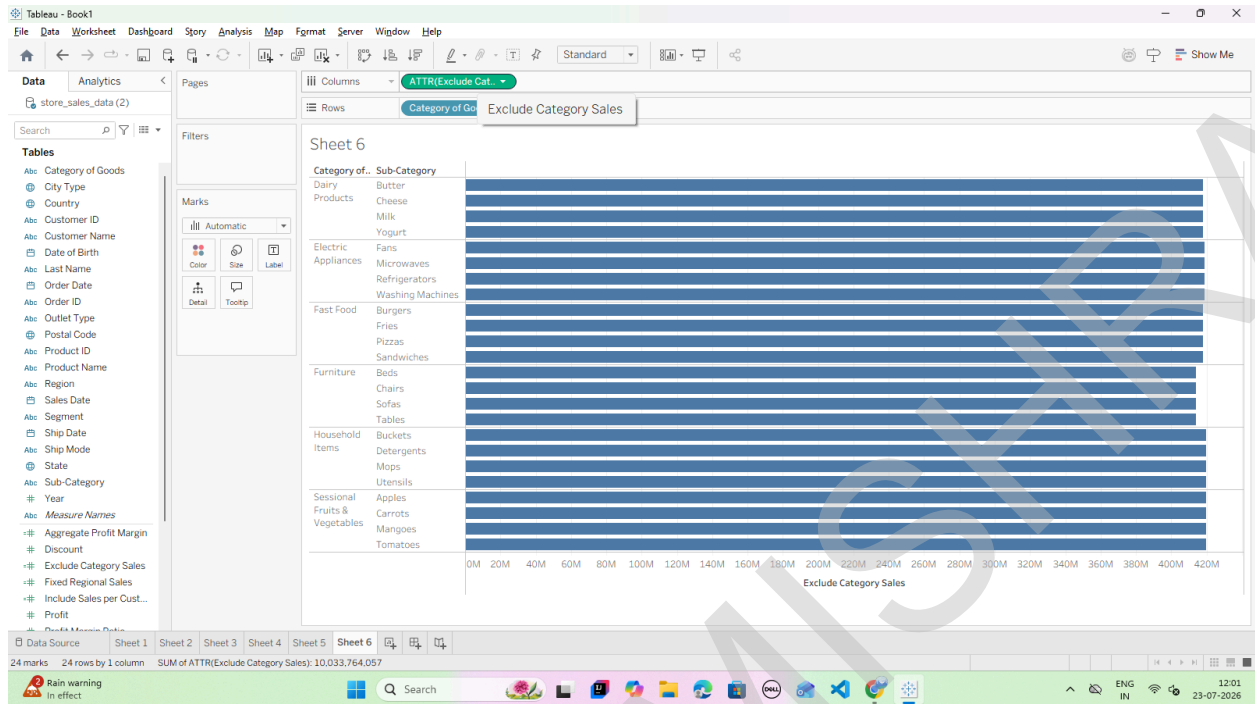
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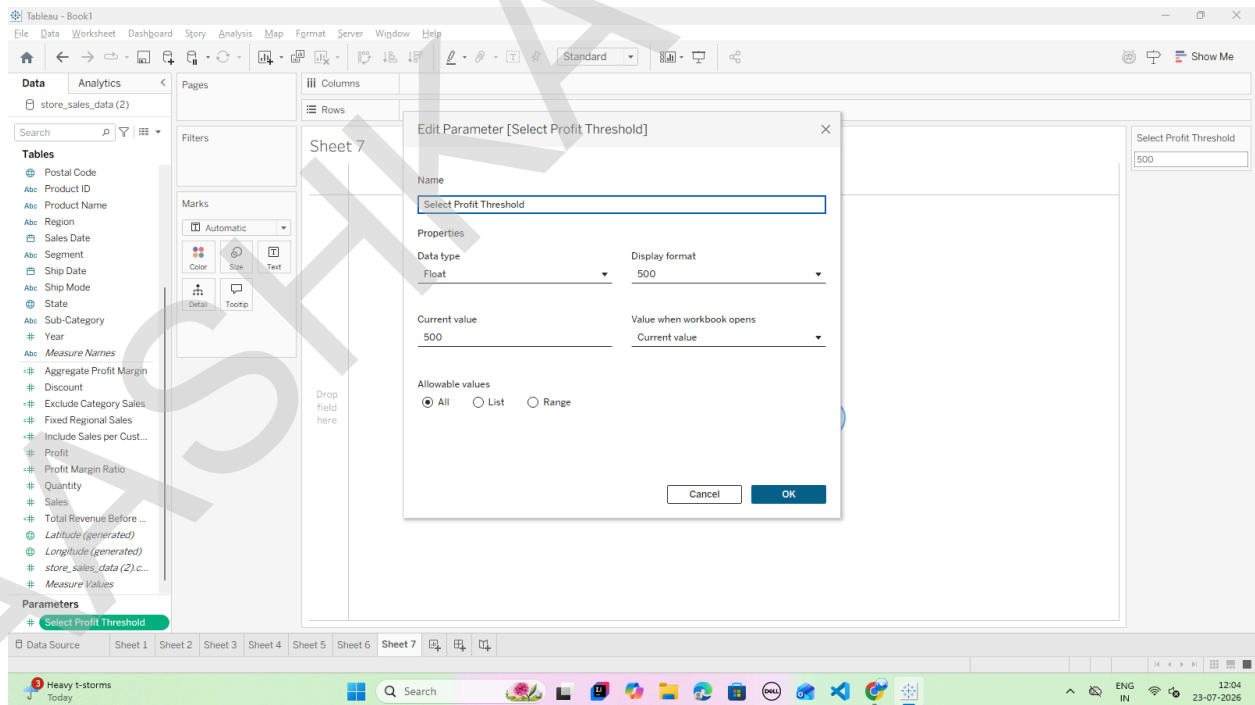
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2. Creating an EXCLUDE LOD Calculated Field



f) Create and use parameters for dynamic analysis

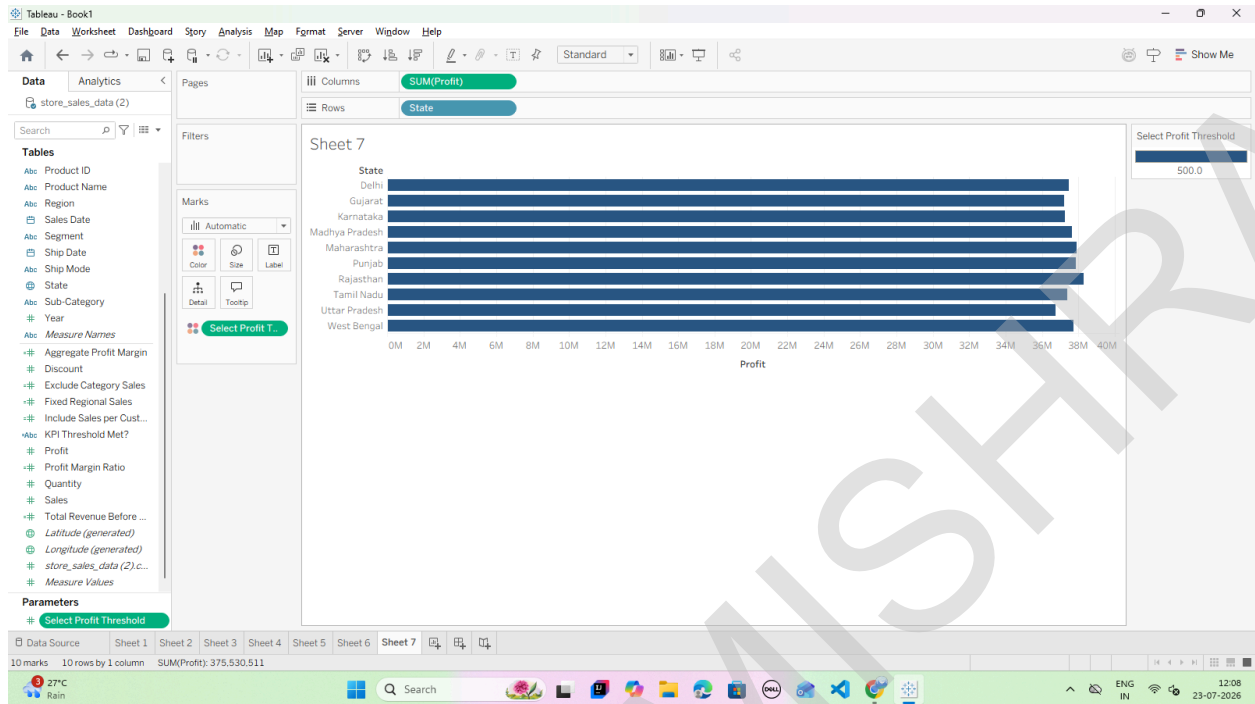
1. Creating a Parameter



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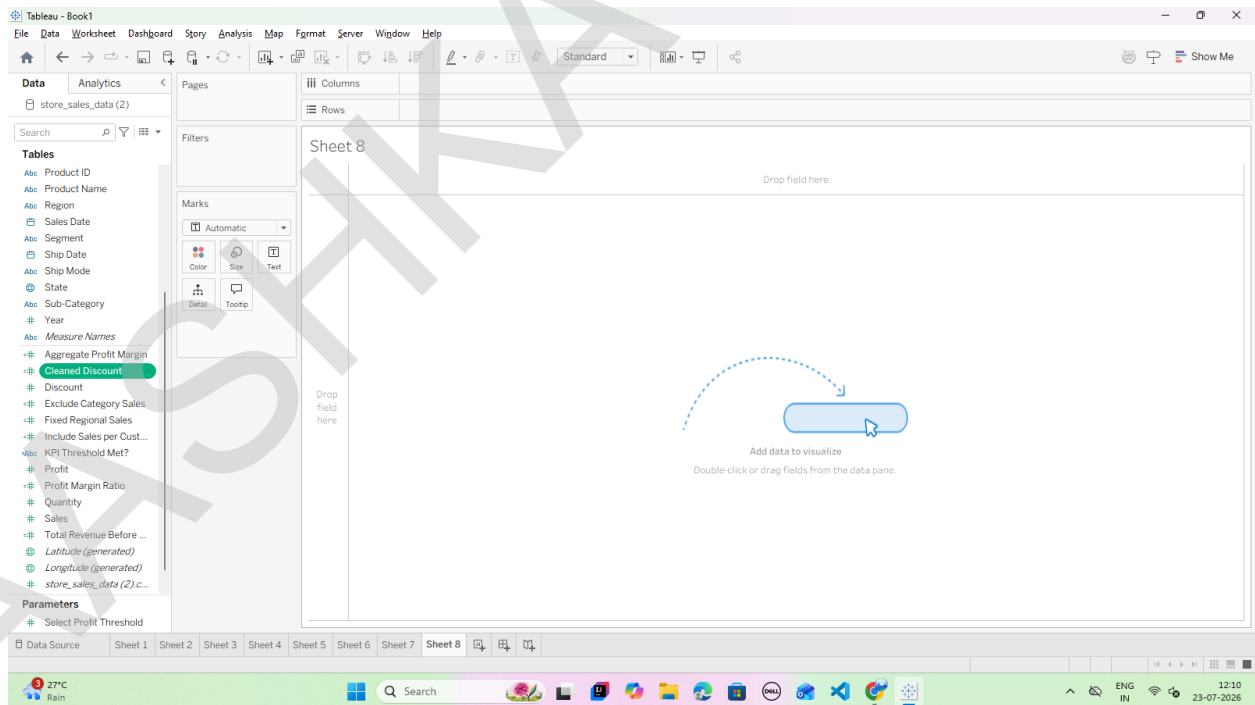
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2. Linking the Parameter to a Calculated Field



g) Apply calculations to fix data issues and enhance insights

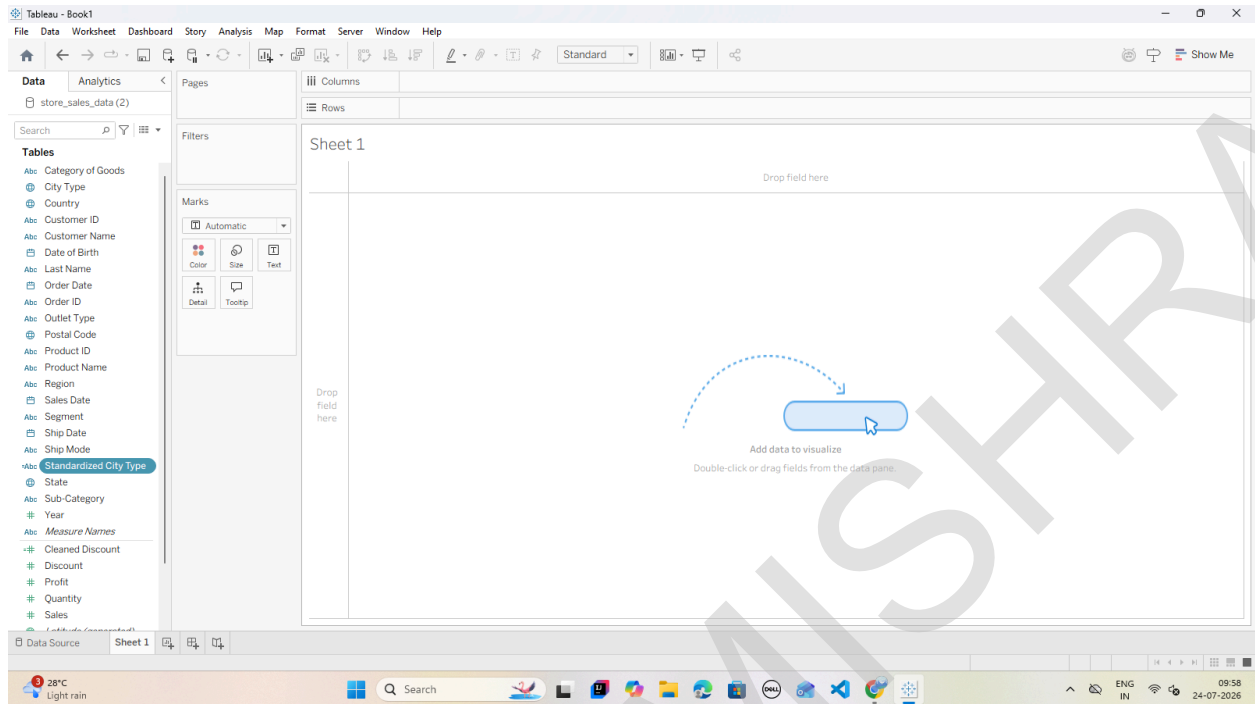
1. Handling Null or Missing Values with IFNULL



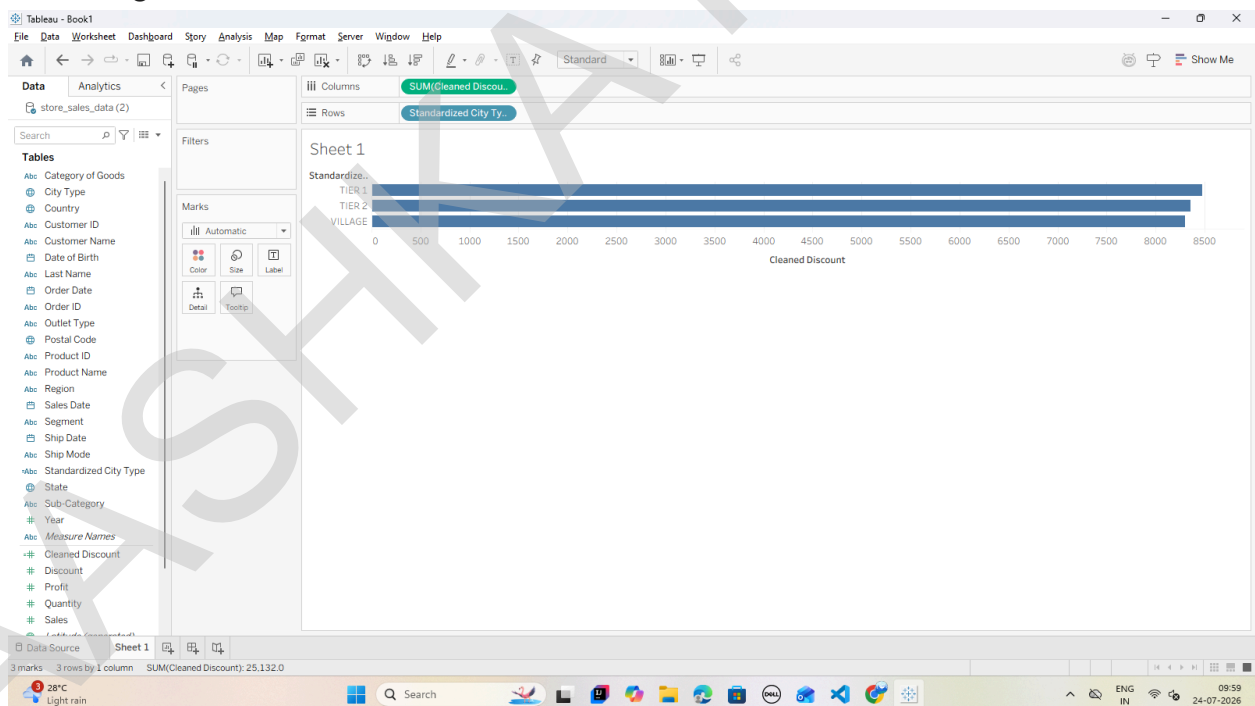
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2. Segmenting Data for Insights using String Functions (UPPER/LOWER)

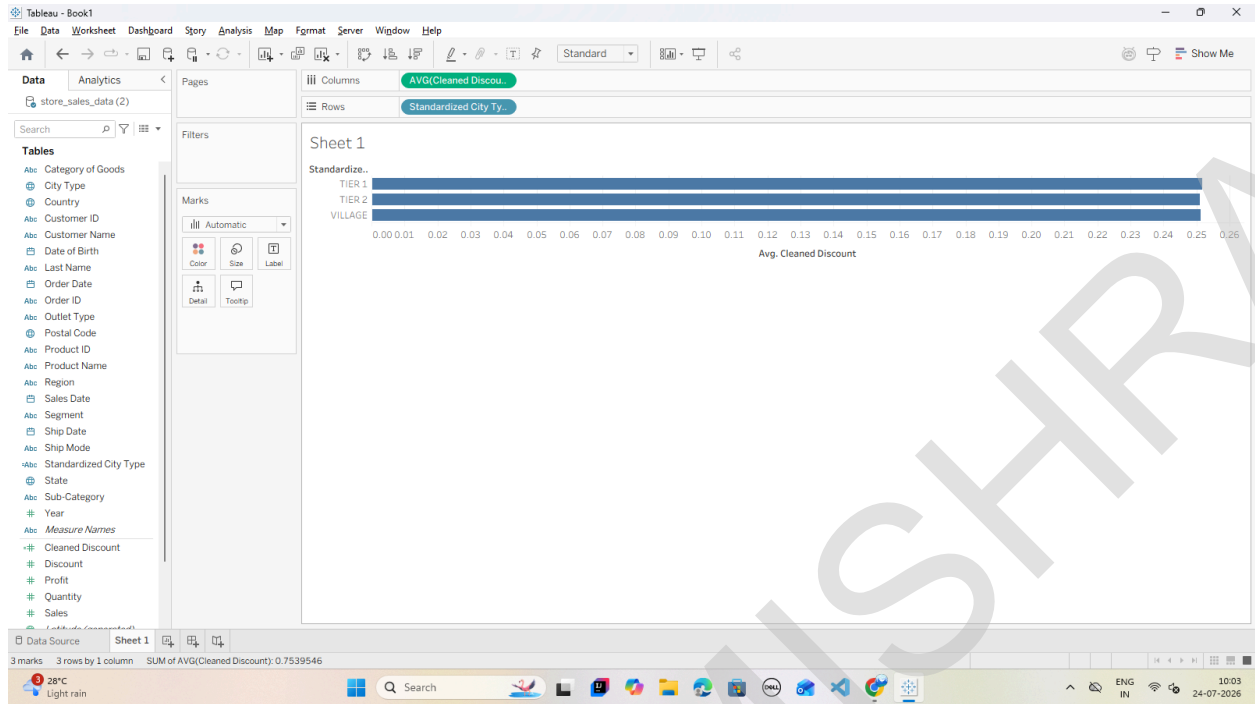


3. Putting the Cleaned Fields to Use



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